

Day 20 Report – DNS Spoofing, SSL Stripping & Defenses

Date: 08 July 2025

Module: DNS Poisoning & SSL Strip Attacks

Mode: Theory + Lab

Main Focus

Explored **DNS spoofing**, **SSL stripping**, and **defensive techniques** to mitigate sniffing.

Theory Topics

- **DNS Spoofing:**
Redirecting DNS queries to malicious IPs
- **SSL Stripping:**
Downgrading HTTPS to HTTP in transit
- **Defensive Measures:**
 - DNSSEC
 - HSTS
 - HTTPS Everywhere

Practical Tasks

- Performed DNS spoofing using `ettercap` plugin
- Used **sslstrip** to downgrade secure sessions
- Verified browser behavior on redirected traffic
- Monitored ARP/DNS logs using `tcpdump`

Key Learnings

- DNS spoofing allows phishing attacks on trusted domains
 - SSL stripping can reveal login credentials on misconfigured sites
 - DNSSEC and forced HTTPS prevent these attacks
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